

CLASS 05 / SEP 22

THE EXCHANGE ATTRACTORS

FIRST HALF: BUILD A CLASSMATE'S

SECOND HALF: DISTANCE DRIVES FORM

MODULE 1 / PARAMETRIC / 1.1 DUE NEXT CLASS

THE EXCHANGE

You will build someone else's object from their words alone.

01

SWAP

Hand your printed procedure to the person opposite. Keep your diagram — they do not get to see it.

02

BUILD

25 minutes. Follow the procedure exactly as written. Do not interpret generously.

03

STOP WHERE IT BREAKS

If a step is ambiguous, stop, write down what was unclear, then make the most literal reading and continue.

04

COMPARE

Put your result next to the author's diagram. The difference is the finding.

DO NOT TALK TO THE AUTHOR. IF YOU CAN ASK THEM, THE PROCEDURE IS NOT BEING TESTED.

WHAT TO RECORD

Both of you write this down. It goes in the booklet.

AS THE BUILDER

- ☐ Every step where you had to guess
- ☐ What you assumed, and why
- ☐ A photo of what you produced

AS THE AUTHOR

- ☐ Which steps were misread, and how
- ☐ What you meant, in one sentence
- ☐ The rewritten step that fixes it

THE USUAL SUSPECTS

Which axis is the cylinder's height along? Does `move()` apply to the centre or a corner? Which way is positive rotation? Is the subtraction the right way round?

AN ATTRACTOR IS A NUMBER

For every point: how far is it from something? Use that number to drive a property.

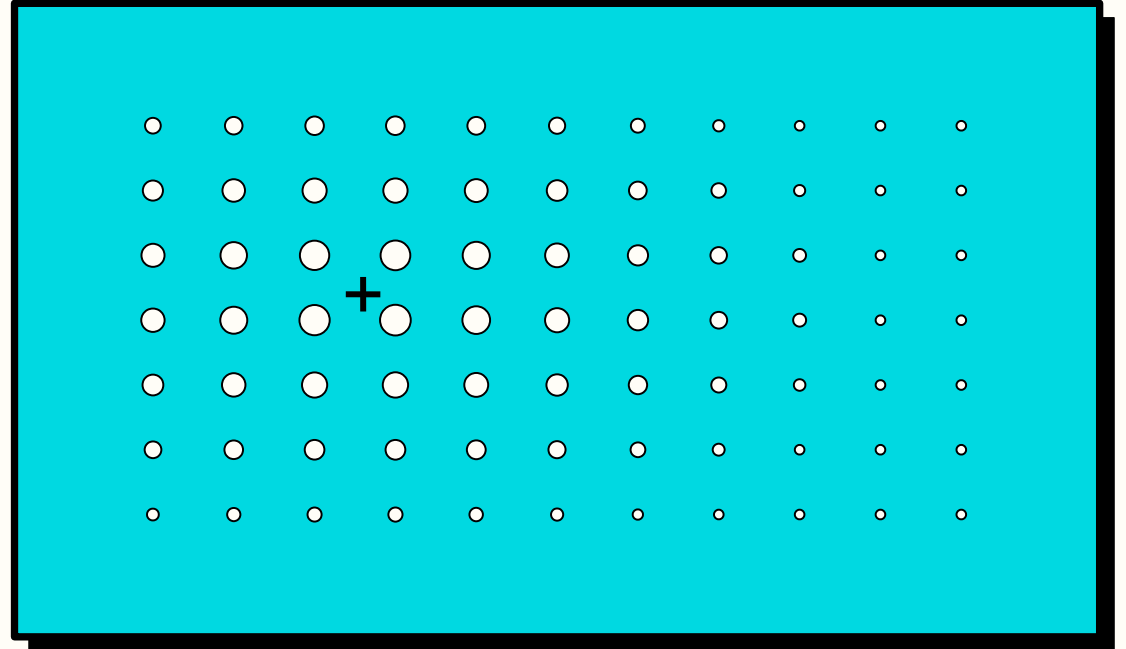
THE WHOLE IDEA

for each point p :

$d = \text{distance}(p, \text{attractor})$

$\text{size} = f(d)$

That is it. Everything else is
a choice of f .

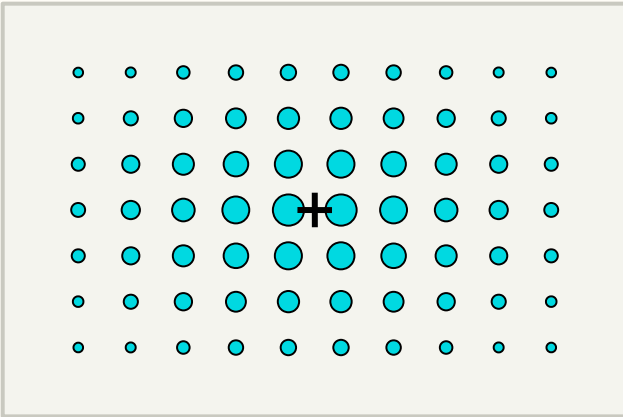


INDEX told you where in the sequence. DISTANCE tells you how far from something. Both are just numbers driving a property — that is the only idea in this module.

THREE KINDS

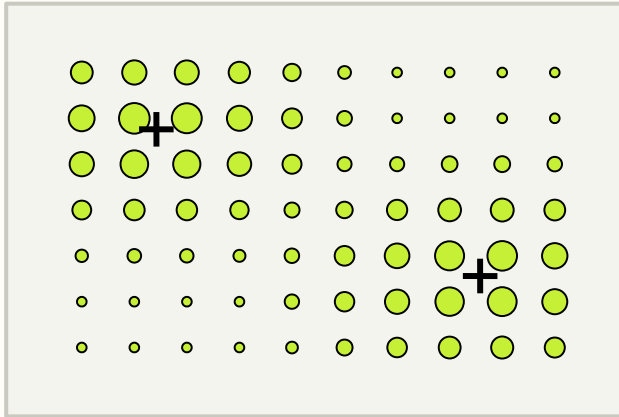
Same maths. What changes is what you measure the distance to.

SINGLE POINT



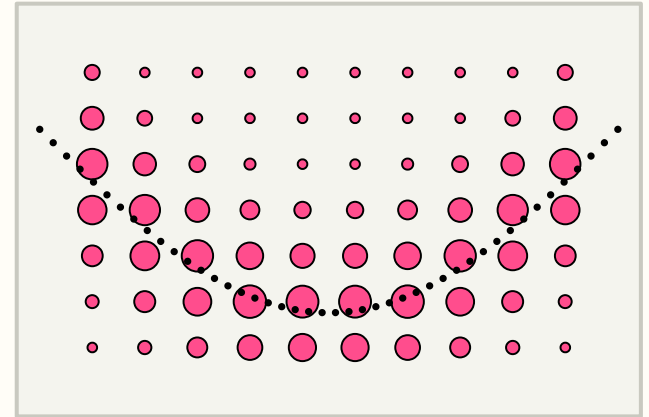
`distance(p, a)`

MULTIPLE POINTS



`min over all attractors`

CURVE

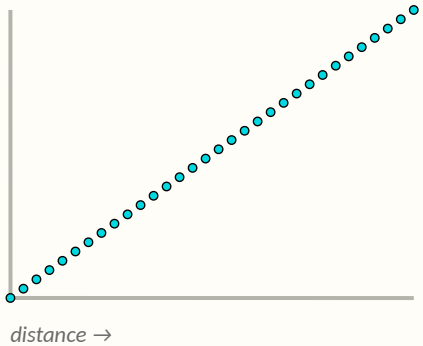


distance to nearest point on the
curve

FALLOFF IS THE FUNCTION

f is your decision. It is where the graphic quality actually comes from.

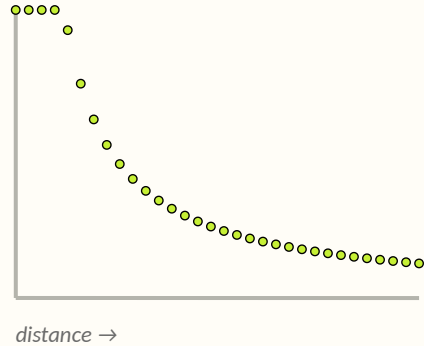
LINEAR



$$v = d$$

Even change all
the way across.

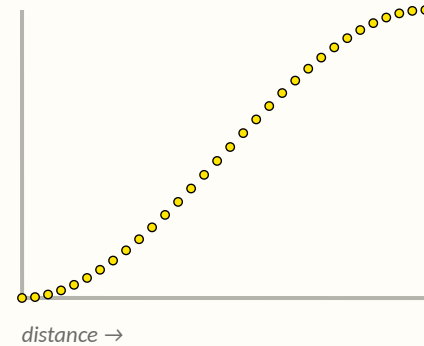
INVERSE



$$v = 1 / d$$

Sharp near the
attractor, flat far off.

SMOOTH



eased

Soft at both ends.
Usually the best-looking.

THRESHOLD



on / off at a cut

No gradient at all.
Two populations.

Two students with the same attractor and different falloff produce completely different images. This is a design decision, not a setting.

REMAP

Distance comes out in model units. Your property needs a specific range. Remap is the bridge.

THE OPERATION

```
t    = (d - dmin) / (dmax - dmin)
out  = outmin + t × (outmax - outmin)
```

```
d      0 ... 8.4 units
out    0.1 ... 1.0 radius
```

SUBTRACT. DIVIDE. MULTIPLY. ADD.

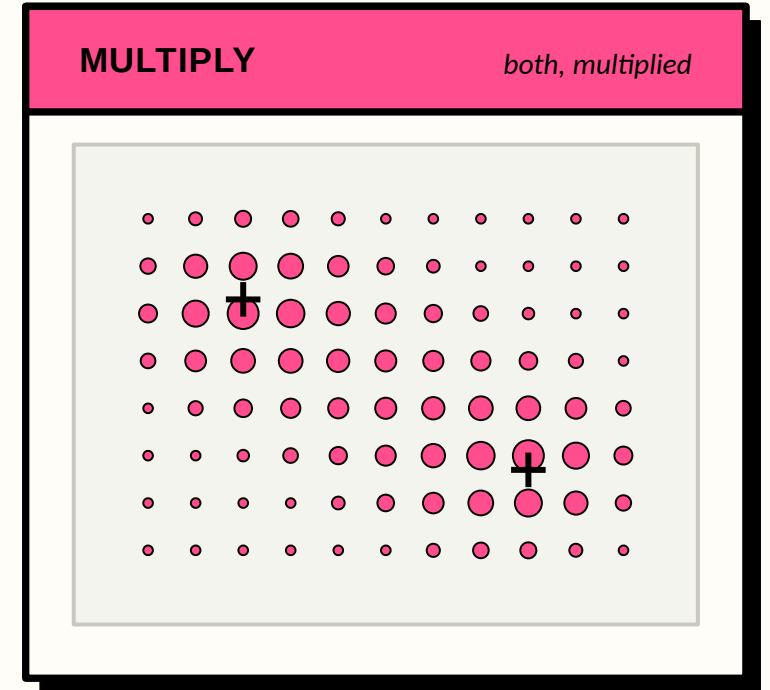
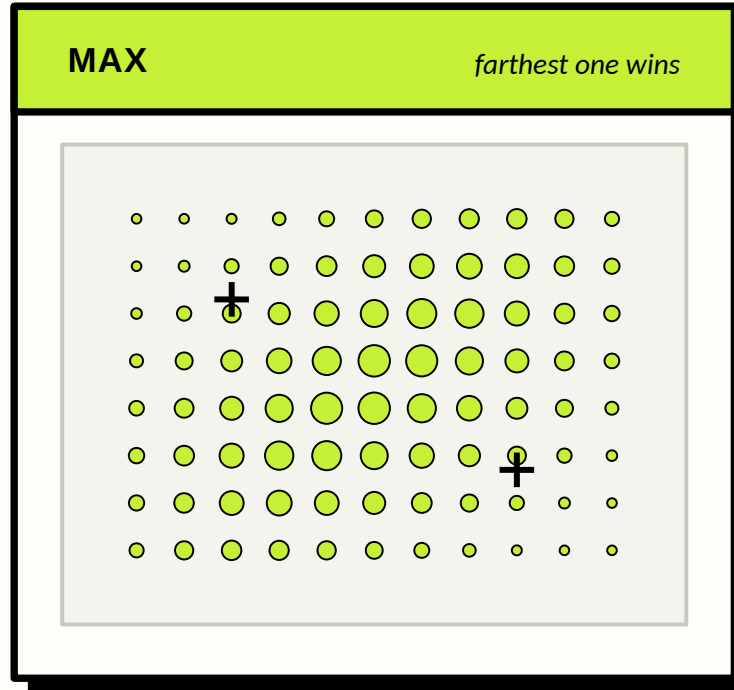
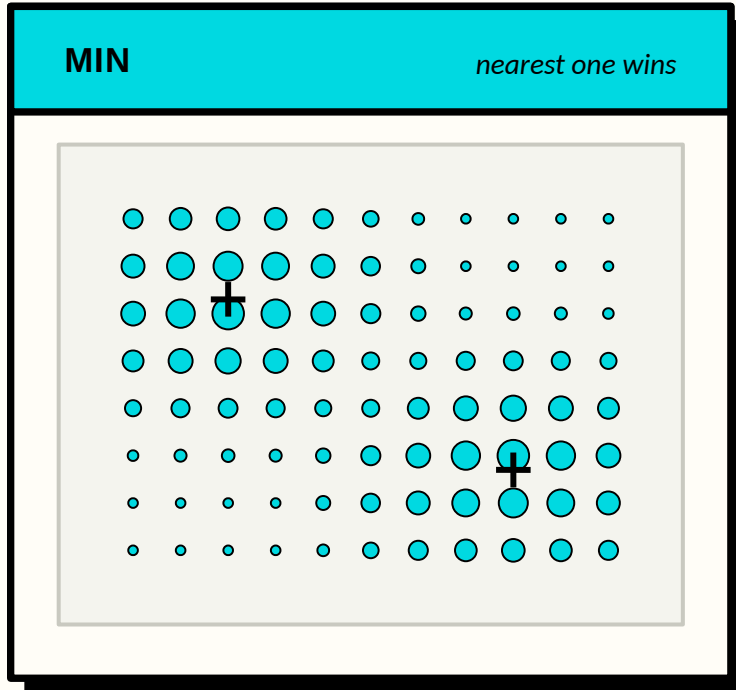
Nothing here is new. It is the arithmetic from Week 2, applied to a distance instead of a coordinate.

Get the ranges wrong and everything is the same size, or one element swallows the sheet.

1.1 REQUIRES YOUR REMAP RANGES DOCUMENTED ON THE SHEET — INPUT RANGE IN, OUTPUT RANGE OUT.

TWO ATTRACTORS, THREE ANSWERS

With more than one attractor each point has a list of distances. How you reduce it is a design choice.



Same two attractors in all three. Only the reduction changed — territories, a single central blob, or two peaks with a pinch between them.

UNDERNEATH, IT IS LISTS

Points, distances, sizes. Three lists, same length, same order.

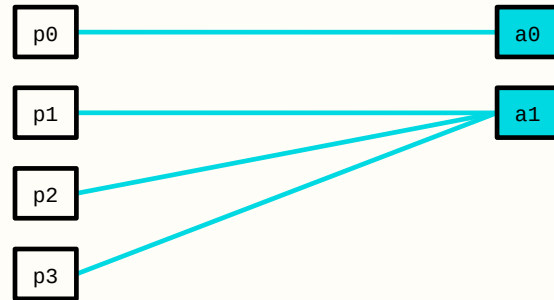
POINTS	DISTANCES	SIZES	PARALLEL LISTS
[0] p_0	[0] 2.1	[0] 0.42	Item [2] in every list describes the same element. Break that correspondence — by sorting or culling one of them — and the sizes attach to the wrong points. That is what “managing lists” means, and it is next week's whole subject.
[1] p_1	[1] 0.4	[1] 0.95	
[2] p_2	[2] 5.8	[2] 0.10	
[3] p_3	[3] 3.2	[3] 0.28	
[4] p_4	[4] 1.0	[4] 0.71	

When a field looks scrambled, it is almost always a list that got reordered somewhere upstream.

WHEN TWO LISTS MEET

4 points, 2 attractors. How do they pair up? The software decides — unless you decide.

LONGEST LIST



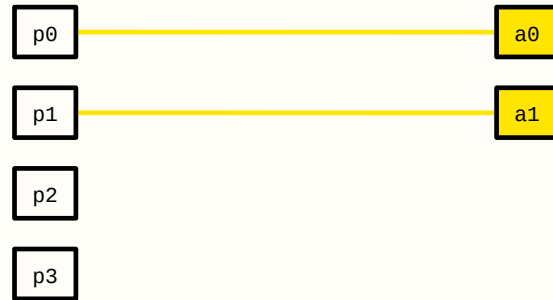
POINTS

ATTRACTORS

4 results

The short list repeats its last item to keep up.

SHORTEST LIST



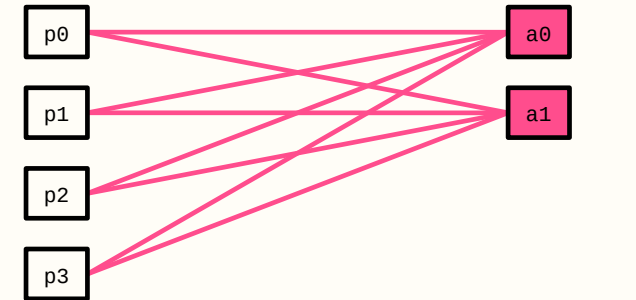
POINTS

ATTRACTORS

2 results

Stops when the short list runs out. Two points get nothing.

CROSS REFERENCE



POINTS

ATTRACTORS

8 results

Every point against every attractor. This is usually what you want.

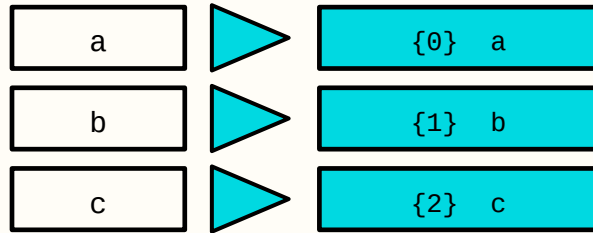
This is why your two-attractor field went wrong. Longest List is the default, and it silently ignored your second attractor for most points.

GRAFT, FLATTEN, WRAP

Three operations that change list structure rather than list contents.

GRAFT

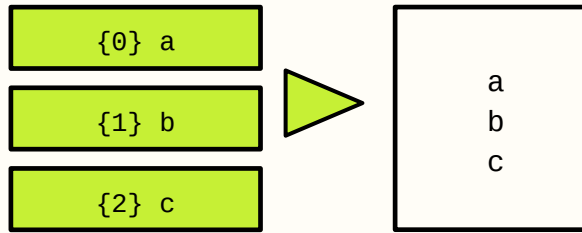
one list → one branch each



Each item becomes its own branch, so the next operation treats them separately instead of pairing them off.

FLATTEN

many branches → one list

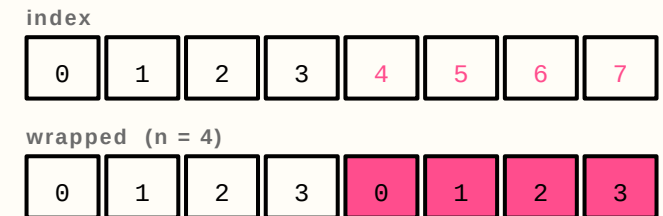


The opposite. Useful at the end — destructive in the middle, because it throws away the structure you built.

WRAP

index past the end → back to the start

$i \bmod n$

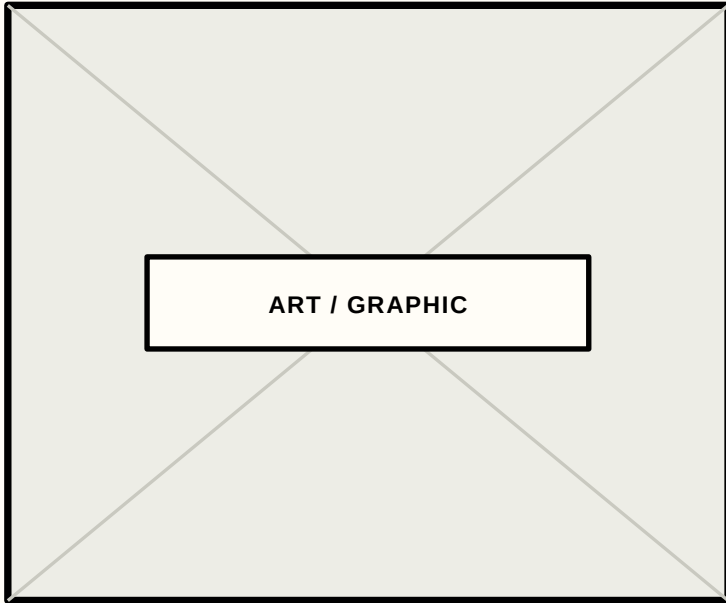


Item 5 of a 4-item list is item 1. This is how you join the last element back to the first.

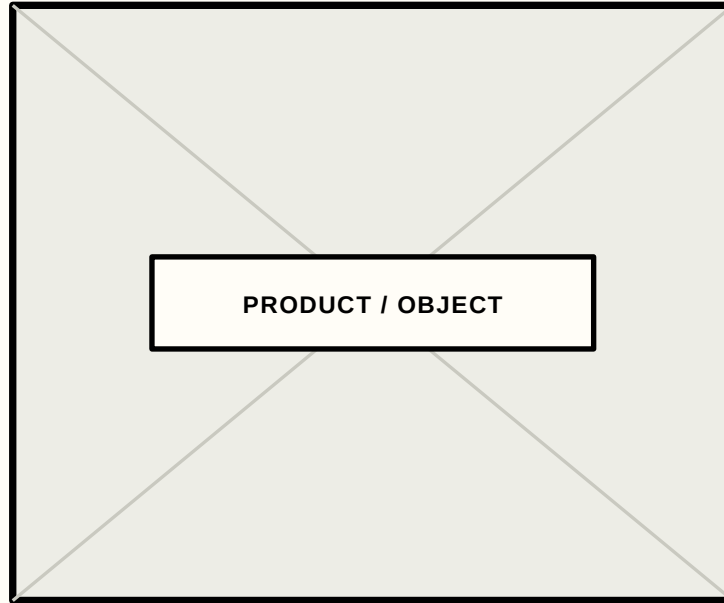
WRAP IS WHY A CONTOUR CLOSES. Connect point $[i]$ to point $[i+1]$ and the last one has no partner — unless the index wraps. That is Module 2 in one line.

ATTRACTORS IN ART, DESIGN, ARCHITECTURE

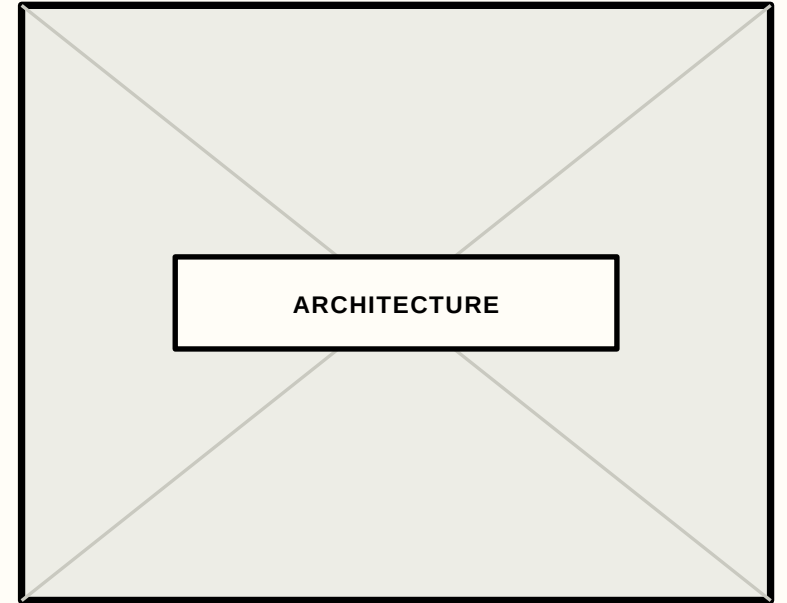
Same three questions: what is the module, what generates the positions, what drives the variation.



Add: a field work — op art, generative graphics, poster.



Add: an object with graded perforation or density.



Add: a façade or screen driven by an attractor logic.

AND ONE MORE QUESTION

Where is the attractor? Sometimes it is a door, a view, a structural point, the sun. Sometimes it is nowhere — chosen purely for the image.

NOW — 1.1 CLINIC

Due next class. Bring your problems to the front.

✓

Three cells of the matrix chosen — which arrays, which attractors

✓

All three attractor types working somewhere in the set

✓

3+ parameters driven by distance — not just scale three times

✓

Remap ranges written down, ready to put on the sheet

✓

One composition over 1,000 elements, one resolved in 3D

✓

Precedent chosen and the three questions answered

THREE DIFFERENT SCALES IS NOT THREE PARAMETERS.

BEFORE NEXT CLASS

FINISH

1.1 is due next class. Two sheets minimum, in your visual identity.

WATCH

Tutorial 2.1 — contouring and list management. Generating contours, then reordering, filtering and reconnecting them.

WRITE

Your mechanism statement for 1.1: how your falloff is computed, in your own words.

FIX

The step in your 0.1 procedure that your partner could not follow. One rewritten line.